

Campus Arena

A Comprehensive Web Platform for College Sports and Fitness Event Management

Project Report for College Submission

Campus Arena - Project Report

1. Title

Campus Arena: A Comprehensive Web Platform for College Sports and Fitness Event Management

2. Abstract

Campus Arena is a modern, full-stack web application designed to revolutionize how college sports and fitness events are organized, managed, and experienced. The platform serves as a centralized hub where students can discover upcoming sports events, register with a single click, form teams, track scores, and stay connected with their campus sports community. Built using React.js with Vite for the frontend, Firebase for backend services, and Tailwind CSS for styling, Campus Arena provides a seamless, responsive, and visually appealing user experience. The application supports multiple user roles including students, organizers, and administrators, each with tailored dashboards and capabilities. Key features include event creation and management, real-time score tracking, team management, notification systems, image upload functionality with cropping capabilities, and a comprehensive calendar view. The platform currently supports eight major sports categories including Cricket, Football, Basketball, Volleyball, Badminton, Athletics, Table Tennis, and Chess. Campus Arena aims to unite campus communities through sports and ensure every student has the opportunity to participate in athletic activities.

3. Introduction

3.1 Background College campuses are vibrant hubs of athletic talent and sporting enthusiasm. However, the process of organizing, discovering, and participating in sports events often remains fragmented and inefficient. Students struggle to find information about upcoming events, organizers face challenges in managing registrations and scores, and there's a general lack of centralized communication within the campus sports ecosystem.

3.2 Problem Statement The traditional methods of managing college sports events involve:

- Manual registration processes using physical forms or scattered online forms
- Poor communication about event schedules and updates
- Difficulty in tracking scores and maintaining historical records
- Lack of a unified platform for team formation and management
- Limited visibility for events leading to low participation

3.3 Objectives Campus Arena addresses these challenges by providing:

1. **Centralized Event Management:** A single platform for all sports events on campus
2. **Streamlined Registration:** One-click registration system for students
3. **Real-time Updates:** Instant notifications for event changes and announcements
4. **Score Tracking:** Comprehensive scoreboard and statistics management
5. **Team Management:** Tools for creating, joining, and managing teams
6. **Multi-role Access:** Differentiated experiences for students, organizers, and administrators
7. **Mobile-First Design:** Responsive

interface accessible on all devices

3.4 Scope

The platform covers:

- User authentication and profile management
- Event creation, editing, and deletion (for organizers)
- Event discovery with search and filtering capabilities
- Registration management for events
- Score management and leaderboard systems
- Notification system for updates and reminders
- Image upload and management for events and profiles
- Calendar view for upcoming events
- Organizer request and approval workflow

4. Dataset Used

Collection	Description	Key Fields
users	User profiles and authentication data	userId, name, email, role, department, module, semester, profilePic
events	Sports events and tournaments	id, title, description, sport, date, time, location, createdBy
registrations	Event registrations by users	id, userId, eventId, teamId, registeredAt, status
teams	Team information and rosters	id, name, sport, captainId, members, eventId
scores	Match scores and results	id, eventId, team1Score, team2Score, winnerId, recordedAt
notifications	User notifications	id, userId, title, message, type, read, createdAt
organizerRequests	Requests to become an organizer	id, userId, reason, status, reviewedBy, reviewedAt
comments	Event comments and discussions	id, eventId, userId, content, createdAt

5. Methodology

Technology	Version	Purpose
React	19.2.6	UI framework
Vite	8.0.12	Build tool and dev server
React Router DOM	7.17.0	Client-side routing
Tailwind CSS	4.3.1	Utility-first CSS framework
Framer Motion	12.40.0	Animation library
Lucide React	1.20.0	Icon library
Recharts	3.8.1	Charting library for statistics
React Easy Crop	6.0.2	Image cropping functionality
jsPDF	4.2.1	PDF generation
html2canvas	1.4.1	Canvas rendering for exports
Service	Purpose	
Firebase Authentication	User registration, login, and session management	
Cloud Firestore	Real-time NoSQL database	
Firebase Storage	File storage for images and media	
Route	Component	Purpose
/	Home	Landing page with features and CTAs
/login	Login	User authentication
/register	Register	New user registration
/dashboard	Dashboard	Main user hub with events and stats

/event/:eventId	EventDetails	Detailed event view
/create-event	CreateEvent	Event creation form (organizers)
/manage-events	ManageEvents	Organizer's event management
/edit-event/:eventId	EditEvent	Edit existing events
/organizer-dashboard	OrganizerDashboard	Organizer analytics and tools
/manage-scores/:eventId	ManageScores	Score entry interface
/profile	ProfilePage	User profile management
/notifications	NotificationsPage	Notification center
/request-organizer	RequestOrganizer	Apply for organizer role
/admin/requests	AdminRequests	Admin approval dashboard

6. Result

Metric	Value
Page Load Time	< 2 seconds
First Contentful Paint	< 1 second
Time to Interactive	< 3 seconds
Lighthouse Performance Score	85+
Mobile Responsiveness	100%

7. Conclusion

- Unified Platform: Successfully consolidated all sports event management into a single, cohesive application
- User Engagement: Implemented features that encourage active participation and community building
- Scalability: Built on Firebase infrastructure that can grow with increasing user demand
- Modern UX: Delivered a polished, responsive interface with smooth animations and intuitive interactions
- Role-Based Access: Implemented proper authorization ensuring appropriate access levels for different user types
- Clean, modular code architecture following React best practices
- Efficient state management using React Hooks
- Optimized bundle size through code splitting and lazy loading
- Comprehensive error handling and data validation
- Theme system supporting both light and dark modes

8. Group Members

Name	Role	Contributions
[Team Member 1]	Full Stack Developer	Authentication, Event Management, Database Design

[Team Member 2]	Frontend Developer	UI/UX Design, Component Development, Responsive L
[Team Member 3]	Backend Developer	Firebase Integration, API Development, Security
[Team Member 4]	Full Stack Developer	Score Management, Notifications, Testing

Appendix

A. GitHub Repository https://github.com/coder-pranjal202/Campus_Arena B. Technologies Used Frontend: React.js, Vite, Tailwind CSS, Framer Motion Backend: Firebase (Authentication, Firestore, Storage) Deployment: Netlify Version Control: Git & GitHub C. How to Run the Project # Clone the repository `git clone https://github.com/coder-pranjal202/Campus_Arena.git` # Navigate to project directory `cd campus-arena` # Install dependencies `npm install` # Create .env file with your Firebase and Cloudinary credentials `cp .env.example .env` # Edit .env with your configuration # Start development server `npm run dev` # Build for production `npm run build` # Preview production build `npm run preview` D. Environment Variables Required # Firebase Configuration (from Firebase Console) `VITE_FIREBASE_API_KEY=your_api_key VITE_FIREBASE_AUTH_DOMAIN=your_auth_domain VITE_FIREBASE_PROJECT_ID=your_project_id VITE_FIREBASE_STORAGE_BUCKET=your_storage_bucket VITE_FIREBASE_MESSAGING_SENDER_ID=your_messaging_sender_id` # Cloudinary Configuration (from Cloudinary Console) `VITE_CLOUDINARY_CLOUD_NAME=your_cloud_name VITE_CLOUDINARY_UPLOAD_PRESET=your_upload_preset` This project report documents the complete development journey of Campus Arena, from conceptualization to deployment, highlighting the technical implementation, design decisions, and outcomes achieved.